

An Explainable Driving Behavior Analytics Framework for Government-Aligned Pay-As-You-Drive Models Using **PiViTel Edge Telemetry**

Shurti Sarmah, Roll no: CSD25007
Jenish Ali Borah, Roll no: CSD25017

1 Background and Policy Context

India is rapid digital transformation in transportation systems under national initiatives such as the **Digital India Mission** [1] and the **Ministry of Road Transport & Highways (MoRTH)** Intelligent Transportation Systems (ITS) programs [4]. These initiatives promote smart mobility, digital governance, and data-driven infrastructure development.

In 2022, the **Insurance Regulatory and Development Authority of India (IR-DAI)** enabled telematics-based motor insurance add-ons including *Pay As You Drive (PAYD)* and *Pay How You Drive (PHYD)* models. Official notification: [2].

These models shift traditional fixed premium insurance structures toward behavior-based and usage-based pricing, aligning with:

- Fair and risk-sensitive premium determination
- Road safety incentivization
- Digital transparency in financial decision-making
- Data-driven governance frameworks

2 Road Safety Motivation

According to the 'Road Accidents in India' report published by MoRTH (2022) [3], over 4.6 lakh road accidents were reported in India, with overspeeding contributing to a significant proportion of fatalities.

This highlights the urgent need for behavior-aware monitoring systems that can:

- Detect risky driving patterns: Unsafe driving behaviours like frequent overspeeding, harsh braking and rapid acceleration.
- Quantify exposure to road hazards: Keeps an eye on where and when the driver drives by measuring the mileage or driving during peak traffic and time spent on accident-prone roads.
- Incentivize safer driving practices: Creates a financial motivation for drivers to improve their driving practices by receiving lower premiums or discounts for the drivers who maintain speed limits, or aggressive driving.

PAYD and PHYD models therefore function not only as insurance innovations but also as behavioral intervention mechanisms supporting national road safety goals.

3 Observational Gap

PiVitel is an edge-based vehicular IoT system designed to collect structured telemetry data including:

- Frame number and timestamp
- Vehicle speed
- Object detection confidence
- Object class
- Distance to detected object
- GPS latitude and longitude
- Spatial object position

While the system efficiently captures high-volume structured CSV data, the following analytical gaps exist:

- Raw telemetry does not directly represent driving quality or risk exposure.
- PAYD frameworks require derived behavioral indicators such as overspeed frequency, sudden acceleration, close-proximity events, and high-risk zone exposure.
- Absence of explainable analytics reduces transparency and stakeholder trust.

Thus, a critical gap exists between IoT data acquisition and policy-aligned driving behavior evaluation.

4 Need for Explainable and Responsible Analytics

Insurance premium determination is a high-stakes financial decision. Therefore, transparency and fairness are essential.

The **NITI Aayog Responsible AI for All** framework [5] emphasizes:

- **Transparency :** The system should be able to explain how and why a particular insurance premium was assigned and see what are the factors influencing the calculation of insurance premium.
- **Accountability:** There must be a clear responsibility for all the decisions made by the systems.
- **Fairness:** Fair premium calculations without be affected by irrelevant or sensitive attributes.
- **Auditability:** Allow independent verification and inspection of the entire process.

Without explainable mechanisms, behavior-based insurance models risk:

- **Algorithmic bias :** Incomplete training data or unbalanced datasets might lead to the model producing systematically unfair outcomes for a group of users.
- **Reduced user trust:** For an innovative system like this, it might take time for the users to gather trust on the system's decisions.
- **Regulatory concerns:** Concerns like system errors, data security, cost and potential misuse of personal data might raise significant concerns in the privacy.

Hence, a structured, interpretable analytics pipeline is necessary for policy-aligned PAYD deployment.

5 Stakeholder and Empathy Analysis

5.1 Drivers

Drivers may experience concerns regarding fairness, privacy, and lack of clarity in score computation. They require transparent dashboards that:

- Explain score derivation
- Highlight risky events
- Provide improvement suggestions

5.2 Insurance Providers

Insurance companies require:

- Reliable behavioral risk metrics
- Fraud detection capability
- Audit-ready scoring mechanisms

5.3 Government and Regulators

Regulatory authorities require:

- Explainable analytics
- Bias mitigation
- Policy-aligned implementation

5.4 Fleet Operators

Fleet managers require comparative analytics, anomaly detection, and safety performance tracking to optimize operations.

6 Problem Statement

The Government of India has enabled behavior-based insurance frameworks such as PAYD and PHYD. These models require accurate, explainable, and transparent assessment of driving behavior.

Although PiVitel generates rich vehicular IoT telemetry in structured CSV format, raw data does not directly provide behavioral risk indicators, safety metrics, or interpretable driving scores required for fair PAYD implementation.

The absence of a structured data analytics and visualization framework limits:

- **Risk quantification** : Raw telematics are unable to transform into meaningful risk scores which makes it difficult the probability of accidents accurately and compare risk levels across drivers.
- **Behavioral abstraction** : Decision-making becomes difficult as the driving signals (speed, braking, acceleration) in the raw data are not able to indicate the driving patterns.

- **Transparency in premium determination** : Lack of explainability of raw data makes it difficult for the user to understand the factors affecting the premium.
- **Stakeholder trust** : Without proper visualization the drivers, insurers and regulators might question the fairness and reliability of the models.
- **Regulatory auditability** : A proper inspection and verification of system is required which is possible only with well-defined analytics and visualization framework.

Therefore, this project proposes the design and implementation of a comprehensive, explainable, and policy-aligned analytics framework that transforms PiViTel telemetry into actionable driving intelligence supporting Government-promoted PAYD models.

7 Research Question

How can raw PiViTel-generated vehicular IoT telemetry be transformed into explainable, risk-sensitive, and policy-aligned driving behavior metrics that support transparent Pay-As-You-Drive insurance models in India?

8 Initial Data Source

The primary data source for this study is the in-house edge device **PiViTel**, developed by **K-Means Technologies Private Limited**. The device generates structured vehicular telemetry data in the form of CSV files during vehicle operation. For the purpose of this project, full access to the PiViTel-generated datasets will be available, allowing comprehensive data processing, analysis, and visualization required to achieve the objectives of the study.

9 Data Analytics and Visualization methods to be used

9.1 Data Analytics Methods

1. **Data Overview**: The objective is to *understand the **structure and composition of the PiViTel telemetry dataset*** before performing analytical processing. Since PiViTel generates large volumes of sensor-based data including **frame numbers, timestamps, vehicle speed, object detection values, detection confidence scores, and GPS coordinates**, it will be necessary to examine the dataset structure in the initial stage. The dataset will be inspected to determine

the **total number of records collected during driving sessions, number of attributes available for analysis, and the data types of each variable such as numerical, categorical, or timestamp values.** Understanding the dataset structure will help identify which variables represent **vehicle motion, environmental interaction, and spatial positioning**, which will be required for behavioural analytics.

2. **Missing Value Analysis:** The objective is to *ensure **data integrity and analytical reliability** before performing behavioural analysis.* PiViTel works on multiple sensors such as cameras, GPS modules, and detection models. During production, sensor interruptions may occur due to environmental conditions, hardware limitations, or signal loss. The dataset will be studied to identify missing entries in variables such as GPS coordinates, object detection values, detection confidence, vehicle speed. Missing values can reduce the reliability of analytical results and affect driving behaviour analysis.

The analysis will include: **calculating missing values for each variable, determining percentage of missing data, deciding whether to remove or handle missing values**

3. **Exploratory Data Analysis (EDA):**

- (a) **Descriptive Statistical Analysis:** The descriptive statistics will be used to summarize the key characteristics of the telemetry dataset. The numerical variables including vehicle speed, object distance and detection confidence will be computed using statistical measures (mean, median, minimum, maximum and standard deviation). These statistics will help to provide an overview of driving behaviour that is being captured in the dataset and establish the baseline conditions for identifying abnormal or risky patterns.
- (b) **Univariate Analysis:** The univariate analysis will be conducted to study the distribution of individual variables independently. Variables in the dataset such as vehicle speed, object distance, and detection confidence is important to be examined using statistical plots and distribution analysis which will help understand the skewness, and spread of the dataset and allow identification of unusual patterns or extreme values.
- (c) **Bivariate Analysis:** The bivariate analysis will be used to examine relationships between pairs of variables in the telemetry dataset for example, relationships between vehicle speed and object distance, vehicle speed and detection confidence, could be analyzed. This analysis will help determine what the certain combination of variables indicate, whether they depict the potentiality of risky

driving behaviour, such as travelling at high speeds while maintaining short distances from detected objects.

- (d) **Correlation Analysis:** Correlation analysis will be performed to determine the strength of the relationships between numerical variables in the dataset. A correlation matrix will be generated to measure the degree of association between attributes vehicle speed, object distance, and detection confidence. This analysis will help in identifying the variables that may significantly influence the driving behaviours of the people and will help in applying the correct feature engineering methods.

9.2 Behavioral Analytics

1. **Feature Engineering:** Feature engineering will be applied to transform raw telemetry variables into meaningful behavioural indicators. Feature engineering becomes important for derived features as they are the ones which are going to provide an interpretable representation of the driving behaviour which will help in behavioural risk analysis. The important features include overspeed events, speed variance, proximity risk indicators, and object encounter frequency.
2. **Risk Metric Computation:** Once the behavioural indicators are computed, risk metrics will be computed based on that that will quantify driving behaviour. These metrics includes overspeed rate, proximity risk score, and speed stability indicators. This evaluation metrics will provide us the level of risk associated with a driver's behaviour based on these telemetry observations.
3. **Driving Score Model** The computed behavioral metrics will give a driving score which represents the safety level of each driver's behaviour. This score will combine behavioural factors such as speed compliance, safe distance maintenance and driving stability to result in a single measure which will further help to study the comparison between different driving ways and provide a proper behaviour-based risk evaluation.

Time-Series Analysis The telemetry dataset mainly contains timestamped observations which will help to perform a time-series analysis to analyze how the driving behaviour changes during a particular driving session. These changes will further allow us to detect the sudden fluctuations or unstable driving patterns, if any.

Anomaly Detection Anomaly detection will be used to identify any unusual driving behaviour observed in the dataset. Anomalies might include extremely high vehicle speeds, sudden proximity to objects or irregular sensor readings which on detecting will help identify the potential risks in the driving behaviour.

Spatial Analysis Spatial analysis will be performed in the geographic coordinates captured by GPS which is contained in the telemetry dataset, to analyze the driving behaviours across different locations. The mapping techniques will be helpful to visualize different vehicle routes and identify the locations where there are chances of risky driving behaviour.

9.3 Visualization Strategy

Visualization helps in understanding and interpreting the PiViTeL telemetry dataset. Since vehicular telemetry data contains multiple sensor measurements such as vehicle speed, object detection, detection confidence, object distance, and GPS coordinates, graphical analysis will be used to explore patterns, identify anomalies, and communicate insights effectively.

In this project, visualization will be performed at multiple stages of the analytical workflow including exploratory data analysis, behavioural risk analysis, and spatial route analysis. A variety of visualization techniques will be used to provide different analytical perspectives of the dataset.

9.3.1 Dataset Overview Visualization

Basic distribution plots (including bar graphs) will be generated to observe the frequency of categorical attributes like detected object classes and observations of sensors. These plots will help in identifying the type of objects frequently detected around the vehicle and provide an overview.

9.3.2 Descriptive Statistics Visualization

The numerical characteristics will be visualized in histogram and density plots which will help to understand the variables vehicle speed, object distance and detection confidence. These plots will help to determine the driving observations whether they are falling within normal ranges or showing extreme behaviour.

The box plots will be used to detect outliers and extreme values in the datasets such as unusually high vehicle speeds or extremely small object distances that may indicate potentially risky driving situations.

9.3.3 Univariate Analysis Visualization

Univariate visualization will include plots such as histogram, density plots and box plots will help in examining the individual telemetry variables independently to analyze the distribution the vehicle speed, object distance and detection confidence by studying the skewness, variability and abnormal patterns if observed in the dataset. Through these

plots, it will be possible to understand the general behaviour of each variable and determine whether the data follows expected patterns.

9.3.4 Bivariate Analysis Visualization

Scatter plots will be helpful in understanding the relationships between different variables between key telemetry attributes to visualize their interactions (scatter plots generated to analyze the relationship between vehicle speed and object distance, and plot between vehicle speed and detection confidence.). These plots help determine whether drivers maintain safe distances while they are traveling at higher speeds and whether environmental conditions influence object detection performance.

Pair plots will be generated to visualize multiple relationships simultaneously, which will provide a broader perspective on how different driving variables interact with one another.

9.3.5 Correlation Visualization

A correlation heatmap will be generated to measure the strength of relationships between numerical variables in the dataset that will visually represent correlation coefficients between variables such as vehicle speed, object distance, and detection confidence.

This visualization will help in identifying variables that may influence driving behaviour and help applying proper feature engineering techniques for behavioural risk analysis.

9.3.6 Behavioural Risk Visualization

To visualize the behavioural risk, bar charts is being used to display the frequency of overspeed events and occurrences of proximity risks based on the behavioural indicators such as overspeed events, proximity risk indicators, speed variance. Histograms will also be generated to visualize the distribution of computed driving scores that will help categorize driving behaviour into different safety levels and provide a clearer interpretation of driver risk profiles.

9.3.7 Temporal Driving Behaviour Visualization

Line plots is used to analyze changes in vehicle speed over time. These plots will help observe how driving behaviour evolves during a driving session and identify parts where aggressive driving behaviour may occur.

9.3.8 Spatial Visualization of Driving Routes

The GPS coordinates allows the spatial visualization helping to analyze geographical driving patterns using mapping libraries such as Folium, vehicle routes can be plotted based on the latitude and longitude data.

Additionally, locations where risky driving behaviour occurs (such as overspeed events) will be highlighted on the map that will allow identification of geographical areas where driving risk may be higher.

9.3.9 Explainable Driving Score Visualization

Finally, visualizations will be used to interpret the driving score model developed in the project. Feature importance plots will be generated to illustrate how different behavioural indicators contribute to the driving score.

These visualizations will support explainable analysis by clearly demonstrating which driving factors most significantly influence driver safety evaluation.

Overall, multiple visualization techniques including histograms, box plots, scatter plots, pair plots, correlation heatmaps, line charts, bar charts, and spatial route maps will be used in this project. The use of diverse visualization methods will enable comprehensive exploration of the telemetry dataset and facilitate effective communication of analytical insights related to driving behaviour and risk evaluation.

References

- [1] Digital India. Digital india mission, 2026.
- [2] Insurance Regulatory and Development Authority of India. Document detail (document id: 394286), 2026.
- [3] Ministry of Road Transport and Highways. Road accidents in india, 2022.
- [4] Ministry of Road Transport and Highways. Intelligent transportation systems (its) programs, 2026.
- [5] NITI Aayog. Responsible ai for all, 2026.